

Facilitator Guide: The National Information Ecosystem

OS-A01-A04 | Candidate version 0.1 | September 2026

Module purpose

Participants learn to see official statistics as a national information ecosystem rather than as the output of one national statistical office. They identify the institutions, data assets, technologies, flows, users, rules, capabilities and risks that determine whether information is fit for public decisions. They then use that map to locate realistic AI opportunities and the dependencies that must be addressed first.

Performance outcome

Given a country scenario or an approved national domain, the participant can produce a scoped ecosystem map with stable identifiers, evidence references, critical flows, user needs, quality and governance risks, coordination gaps and a justified statement of where AI may or may not add value.

Module	A1
Competencies	C1 Statistical purpose and public value; C3 stewardship; C4 quality; C6 governance; C8 communication.
Entry level	No prior AI expertise required; basic familiarity with a statistical organization or sector.
Exit level	L1 guided practitioner, with an optional L2 workplace assignment.
Structured time	6 hours, delivered as one day or two 3-hour sessions.
Practice share	Approximately 65 percent.
Primary evidence	Completed ecosystem map, flow analysis, risk statement and oral defence.
Phase 1 link	Uses the concepts and record types of Phase 1D.

Package contents

Part	Use
Use	Controlled facilitator edition. Do not distribute answer and scoring guidance before learner activities are complete.

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Learning outcomes and evidence

By the end participants can	Evidence
Define the boundary of a national or sector information ecosystem.	Scope statement naming domain, geography, period, users and exclusions.
Distinguish producer, source owner, steward, processor, coordinator, funder, regulator and user roles.	Actor register with role and authority evidence.
Inventory data assets from surveys, censuses, administrative records, research, geospatial and private sources.	Asset register with provenance and quality notes.
Trace one critical information flow from collection or generation to use.	Flow record showing transformations, handoffs, delays and failure points.
Connect user needs to fitness-for-use requirements.	Need-to-asset matrix with unmet or partially met needs.
Identify coordination, capability, quality, legal, security and sustainability risks.	Prioritized gap and risk list.
Locate bounded AI opportunities without assuming AI is the answer.	Opportunity and dependency statement with a proceed, investigate or do not pursue recommendation.

Critical safeguards

- Do not include confidential records, passwords, security details or personal data in the classroom map.
- Label claims as evidenced, reported or unknown. Do not convert informed opinion into fact.
- Map actual decision authority; do not assume the NSO controls every sector source.
- An AI opportunity is not selected until purpose, data access, quality, human responsibility and security are considered.

Facilitator preparation

Before delivery	Action
Two weeks before	Confirm participants and roles; choose fictional case only or add an approved national application.
One week before	Collect non-sensitive organization charts, statistical programmes, sector plans and public metadata if using a country application.
Three days before	Prepare printed or digital actor, asset, flow, need, risk and opportunity templates.
One day before	Test display, collaboration tools and offline fallback; assign mixed-role groups of four to six.
At opening	State evidence, confidentiality and respectful-challenge rules; explain that unknown is an acceptable finding.

Room and materials

- Wall or digital workspace with six zones: actors, assets, flows, needs, risks and opportunities.
- One case packet per participant and one mapping pack per group.
- Colored notes or equivalent labels for evidenced, reported and unknown claims.
- Timer, scoring rubric and facilitator observation sheet.
- Optional country documents that have been cleared for training use.

Facilitator stance

Do not reward the longest inventory. Press groups to define boundaries, cite evidence, distinguish formal authority from practice, identify missing actors and follow information through transformations to a real user decision. The most valuable finding may be an unknown, an unowned handoff or a need that no current asset can satisfy.

Delivery plan

Time	Activity	Participant output	Facilitator focus
00:00–00:20	Baseline challenge	Initial sketch of who produces agricultural production information.	Expose NSO-only assumptions without correcting immediately.
00:20–00:45	Concept clinic	Annotated definition and ecosystem components.	Clarify system boundary, roles, assets, flows and users.
00:45–01:05	Demonstration	Observation checklist from a worked mini-map.	Model evidence labels, identifiers and unknowns.
01:05–01:50	Exercise 1 actors and authority	Actor register.	Challenge omissions and duplicated mandates.
01:50–02:00	Break	—	Review group progress.
02:00–02:45	Exercise 2 assets and sources	Asset register.	Separate asset, technology and publication.
02:45–03:00	Synthesis	First dependency statement.	Connect actors to assets.
03:00–03:20	Recap and challenge variation	Corrected baseline sketch.	Surface changes in participant reasoning.
03:20–04:20	Exercise 3 critical flow	End-to-end flow and failure analysis.	Follow transformations and decision points.
04:20–04:30	Break	—	Select issues for debrief.
04:30–05:10	Exercise 4 needs risks and AI	Need matrix and opportunity screen.	Prevent technology-first proposals.
05:10–05:40	Group defence	Five-minute evidence-based presentation.	Use challenge questions and rubric.
05:40–05:55	Individual assessment	One-page scoped ecosystem brief.	Assess individual competence.
05:55–06:00	Transfer commitment	Workplace assignment and owner.	Confirm authorization and follow-up.

Answer and decision guide

Exercise 1 reference reasoning

- ACT-01 has evidenced national coordination authority under EVD-01 and produces the annual survey. Its authority over sector release methods requires interpretation of designation arrangements, so some detail remains unknown.
- ACT-02 is a sector producer; the unfulfilled EVD-02 handoff shows a coordination and compliance problem rather than proof that its statistics are invalid.
- ACT-03 owns or generates extension reports but its proximity-to-road coverage creates selection risk.
- ACT-06 is a high-consequence user and integrator; it is not automatically the methodological authority for crop statistics.
- ACT-07 and ACT-08 are research producers whose evidence requires external-validity and sustainability assessment.
- A likely missing actor is the data-protection or security authority, treasury, regional administration, private imagery or platform provider, or a clearly named farmer subgroup. Defensible alternatives are accepted.

Exercise 2 reference assets

Asset	Fitness issue
Agricultural census and holding frame	Coverage may be obsolete between censuses.
Annual production survey	Potentially stronger statistical design but too late for early alerts.
Extension crop-condition forms	Timely but regionally inconsistent and coverage-biased.
Registered parcels	Customary holdings missing; legal parcel is not necessarily operational holding.
Weather observations	Missing station periods and spatial representativeness.
Food-security integration database or report	Depends on coherence, timing and transparent treatment of conflicting inputs.
Research trial data	Controlled or selected conditions limit population generalization.
Experimental satellite crop map	Validation narrow in geography; funding and staffing fragile.
Market price series	Selected markets and telephone collection may affect coverage and continuity.
Classifications geographies and identifiers	Critical missing integration assets under EVD-07.

Exercise 3 reference conclusion

Strong answers usually identify definition and revision incoherence between ACT-01 and ACT-02; biased and inconsistent extension reporting; missing identifiers; the six-week timing mismatch;

narrow satellite validation; or fragile geospatial capability. The response must connect the chosen failure to the cabinet decision and name a responsible role. A reasonable non-AI priority is a harmonized crop and season definition, common revision status, documented early-estimate protocol, benchmark sample, flow calendar or minimum metadata exchange.

Exercise 4 decision

“Automatically produce estimates” is not ready for a pilot. A defensible decision is investigate dependencies first and narrow the use case. Candidate bounded uses include assisting analysts to document discrepancies, flagging missing metadata, summarizing reviewed field reports or testing satellite classification for selected crops and districts against an independent benchmark.

Generative AI should not be the estimator or release authority. Proceed only after ownership, definitions, benchmark, access, review, security, recurrent resources and fallback are specified.

Assessment rubric

Criterion	Points	Competent evidence
Boundary and public purpose	15	Domain, product, user decision, geography, period and exclusions are clear.
Actors roles and authority	15	Critical actors and distinct roles are correctly identified; authority is not assumed.
Assets and provenance	15	Multiple source types are inventoried with owners and material fitness issues.
Flow and dependencies	15	Transformations, handoffs, controls, responsibilities and failure points are traceable.
Evidence discipline	10	Material claims use E, R or U labels and references; unknowns remain visible.
Fitness for use	15	Quality requirements follow from the user decision and consequences.
AI judgement	10	Recommendation is bounded, dependency-aware and does not bypass professional authority.
Communication	5	Map and conclusion are concise, internally consistent and defensible.

Decision rules

- Competent: 70 or more, at least half the available points in every criterion and no critical failure.
- Lead-level performance: 85 or more plus effective challenge of conflicting evidence and a defensible prioritization decision.

- Critical failures: inclusion of protected data without authorization; fabricated evidence; assigning release authority to an AI system; or recommending integration while concealing material incompatibility.
- Reassessment: correct the failed map elements and complete a new individual boundary-and-decision brief.

Technical validation and release record

Check	Version 1 result
Statistical concepts	Aligned to Phase 1D ecosystem records and Phase 2A competencies.
Case coherence	Actor and evidence IDs reconcile; contradictions are deliberate and documented.
Answer logic	Allows defensible alternatives while preserving critical safeguards.
Assessment	Individual evidence separates group performance from participant competence.
Confidentiality	Fictional case contains no personal or confidential data.
Tool dependency	None; can be delivered on paper, offline or with common collaboration tools.
Accessibility	Tables use explicit headers; exercises have text alternatives and do not depend on color alone.
Localization	Fictional names and country conditions can be replaced without changing core outcomes.

Release role	Required sign-off before national use
Product owner	Confirms scope, resources and next review.
Statistical reviewer	Confirms ecosystem concepts, source types and fitness reasoning.
Domain reviewer	Confirms agricultural realism when used as the main example.
Safety reviewer	Confirms approved-data and tool rules for any local exercise.
Learning reviewer	Observes a pilot delivery and calibrates the rubric.

Version record

Version	Date	Status	Next action
1.0	September 2026	Reference teaching package ready for controlled pilot.	Pilot with a mixed-role cohort; record timing, scoring differences, unclear evidence and localization needs.

Next package

After the controlled pilot and revision of A1, develop Block A2 AI Concepts for Statisticians using the same production standard. A2 should begin with task and failure classification, not a historical tour of AI or a vendor demonstration.